

## Chapter-1

1. Define statistics as a subject and explore its importance in CSE.
2. Define Variable with examples. What is the difference between discrete and continuous variables? Identify whether these variables are discrete/continuous/qualitative?  
The temperature of a CPU or a server, The amount of memory being used by a process, Program execution time, The count of clicks on a IIUC webpage, The programming language being used in a project, etc

## Chapter-2

3. Explain the procedure for calculating the geometric mean, median, and mode for group and ungrouped data using illustrative examples. Write down the formulae for computing quartile, decile, and percentile from a grouped frequency distribution.
4. A database administrator is monitoring the response times of queries in a database system. The response times are grouped into intervals (in milliseconds) as shown in the frequency table below:

| Response Time (ms) | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 |
|--------------------|------|-------|-------|-------|-------|
| Frequency          | 3    | 8     | 12    | 5     | 2     |

Calculate (i) the mean, median and median response time for the queries (ii) 1<sup>st</sup> quartiles, 6<sup>th</sup> deciles and 78<sup>th</sup> percentiles

5. In a software development team, the time (in hours) taken to complete code reviews was recorded: 2, 4 and 8  
Using the data justify: (i)  $A.M \geq G.M \geq H.M$  (ii)  $A.M \times H.M = GM^2$

## Chapter-3

6. Define mean deviation, variance, standard deviation and coefficient of variation (CV).
7. A software company is assessing the following performance of two programmers, Shafeen and Shahariar in terms of the number of lines of code they write per hour.  
Shafeen 's coding speed over five days: 130, 135, 125, 140, 150  
Shahariar 's coding speed over five days: 120, 125, 130, 135, 150  
Calculate the coefficient of variation for both Shafeen and Shahariar and determine who has a more consistent coding speed.